INCH-POUND

MIL-DTL-32173
20 August 2004

DETAILED SPECIFICATION

GLOVES, FLYER'S, INTERMEDIATE COLD, HAU-15/P

1. SCOPE

1.1 Scope. This document covers one type of intermediate cold flyer's gloves, HAU-15/P.

1.2 Classification. The gloves will be of one type as follows.

1.3 Schedule of sizes The gloves will be of the following sizes (see 6.2).

SCHEDULE OF SIZES
5, 6, 7, 8, 9, 10 and 11

Comments, suggestions, or questions on this document should be addressed to: Defense Supply Center Philadelphia, Clothing and Textiles Directorate, Attn: DSCP-COCT (Bldg 6), 700 Robbins Ave., Philadelphia, PA. 19111-5092 or emailed to Leo.Bailey@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at www.dodssp.daps.mil.

AMSC N/A **FSC 8415**

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

FEDERAL SPECIFICATIONS

V-T-295 - THREAD, NYLON

COMMERICAL ITEM DESCRIPTIONS

A-A-50195 - THREAD, ARAMID
A-A-50199 - THREAD, POLYESTER CORE, COTTON OR POLYESTER-COVERED

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-C-3735 - CUFFS, KNIT, WRIST & ANKLE AND CLOTH, KNITTED

MIL-W-5664 - WEBBING, TEXTILE, ELASTIC

MIL-DTL-32067 - LEATHER, SHEEPSKIN, CHROME TANNED

MIL-DTL-32075 - LABEL: FOR CLOTHING, EQUIPAGE, AND TENTAGE, (GENERAL USE)

MIL-B-44346 - BATTING, SYNTHETIC FIBER, MEDIUM DENSITY

MIL-C-81393 - CLOTH, KNITTED, POLYAMIDE, HIGH TEMPERATURE RESISTANT SIMPLEX, JERSEY

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/or www.dodssp.daps.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

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2.2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

"ASTM INTERNATIONAL"

ASTM-D-1388	Standard Test Method for stiffness of Fabrics
ASTM-D-1777	Standard Test Method for Thickness of Textile Materials
ASTM-D-2724	Standard Test Method for Bonded, Fused, and Laminated Apparel Fabrics
ASTM-D-3776	Standard Test Method for Mass Per Unit Area (Weight) of Fabric
ASTM-D-3787	Standard Test Method for Bursting Strength of Textiles-Constant-Rate-of -Traverse (CRT) Ball Burst Test
ASTM-D-3887	Standard Test Method for tolerances for Knitted Fabrics
ASTM-D-4075	Standard Test Method for Compounding Materials Flame Atomic Absorption Technique-Determination of Metals
ASTM-D-5053	Standard Test Method for Colorfastness of Crocking of Leather

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or www.astm.org or from the ASTM International, 100 Barr Harbor drive, west Conshohocken, PA 19428-2959.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQC Z1.4	-Sampling Procedures and tables For Inspection by Attributes

(Copies of these documents are available online at http://www.ansi.org or from the American Society for Quality Control, 611 East Wisconsin Avenue, Milwaukee, WI 53203.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First articles. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2 (see 6.3.)

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3.2 Recycled, recovered, or environmentally preferable materials Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Basic material.

3.3.1 Leather. The leather of the palm, finger palm, palm patch and welt components of the shell shall conform to Type A, Class 2, Subclass B of MIL-DTL-32067 except the leather shall have a stitch tear of not less than 12 pounds and the requirements for the total ash, chromic oxide, and pH shall not apply. The leather shall be drum dyed black and the color shall be uniform throughout the grain surface. Only tap water shall be used for dampening the leather prior to cutting. (see 4.4.1).

3.3.2 Cloth . The knitted cloth for the remainder of the shell shall conform to MIL-C-81393. The color of the cloth shall be Sage Green, USAF shade No. 1565 (see 4.4.1).

3.3.3 Lining. There shall be two types of linings used in the construction of this glove, the difference being the insulation used in the lining. Lining A is used for the hand back, thumb back and finger back. Lining B is used for the hand palm and finger palm. Lining A and Lining B insulation shall be adhered to the same backing and facing fabrics.

3.3.3.1 Backing and facing. The backing fabric shall be 2-bar tricot knitted with 15 denier nylon monofilament fiber. The facing fabric shall be 100% cotton jersey knit. The fabrics shall meet the physical requirements specified in Table I when tested as specified in 4.4.1

TABLE I. Physical requirements of backing and facing fabrics

Characteristic	Requirement	
Color	Backing	Facing
Weight, oz/sq yd:	Natural	White
min	0.4	3.2

TABLE I. Physical requirements of backing and facing fabrics (con't)

Characteristic	Requirement	
Max	0.8	3.2
Fabric count:		
Wales/inch (min)	20	30
Courses/inch min	20	40
Bursting strength lbs (min)	20	49

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3.3.3.2 Polyester insulation for Lining A. The insulation to be adhered to the backing and facing fabrics shall be a polyester batting meeting the requirements of MIL-B-44346, except that the thickness and weight shall be as specified in Table II when tested as specified in 4.4.1.

TABLE II. Physical requirements of Lining A Insulation.

Characteristic	Requirement Minimum	Maximum
Thickness, inches	.25	.35
Weight, oz/sq yd	2.7	3.3

3.3.3.3 Polyester insulation for Lining B. The insulation to be adhered to the backing and facing fabrics shall be a polyester batting, 3M style Thinsulate Type C-40, 40 gram weight.

3.3.3.4 Finished lining, A and B. The finished lining (nylon backing/polyester batting/cotton facing laminate) shall be constructed by bonding the technical back of the nylon backing fabric to one side of the polyester batting and bonding the technical back of the white cotton facing fabric to the other side of the polyester batting. The finishing lining laminates shall meet the requirements of Table III when tested as specified in 4.4.1 (see 6.5.1).

TABLE III. Physical requirements of finished linings.

Characteristic	Requirement Minimum	Maximum
Thickness, Lining A, inches	0.18	0.30
Thickness, Lining B, inches	0.10	0.22
Drape Stiffness, Lining A, inches		3.53
Drape Stiffness, Lining B, inches		2.9
Bond Strength, Lining A, lbs/3inch width		
Insulation to facing	0.96	1.3
Insulation to backing	0.786	1.064
Bond Strength, Lining B, lbs/3 inch width		
Insulation to facing	0.11	0.15
Insulation to backing	0.081	0.109

3.3.4 Knit Cuff. The knit cuff shall conform to Type IV, class 2 of MIL-C-3735. except that the cuff shall be 7 inches in length, folded to 3 1/2 inches, flared for approximately 1 1/2 inches at each end. The color of the cuff shall be Sage Green, USAF 1535 (see 4.4.1, 6.5.2)

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3.3.5 Elastic . The elastic for the wrist and back of hand take-ups shall be 1/2 inch wide cotton conforming to Type I, Class 1, of MIL-W-5664. The lengths of elastic to be used for each size glove shall be as listed in Table IV. The same length of elastic shall be used on the dorsal side (back) of the hand across the wrist and across the palm. (see 4.4.1)

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TABLE IV. Length of elastic take-ups. In Inches

Size	Palmar side at wrist across palm/thumb Back assembly	Dorsal side at palm And wrist	Tolerance
5	4	3 5/8	+1/4, -0
6	4 1/8	3 3/4	+1/4, -0
7	4 1/4	4	+1/4, -0
8	4 1/4	4	+1/4, -0
9	4 5/8	4 1/8	+1/4, -0
10	4 5/8	4 1/8	+1/4, -0
11	4 7/8	4 1/4	+1/4, -0

3.3.6 Thread The thread for stitching the gloves shell and hemming the gloves shall be a black 400 Denier or 40 Tex aramid thread conforming to A-A-50195. The thread for stitching the lining and attaching the knit cuff to the lining assembly shall be natural nylon conforming to Type I or II, size B of V-T-295. As an alternate, a 36 to 45 Tex cotton-covered polyester core thread conforming to A-A-50199, Type I may be used for stitching the lining (see 4.4.1).

3.4 Label. Each glove shall have a combination size and identification label conforming to type VI, Class 4, of MIL-DTL-32075. The label shall show fastness to laundering transference as specified therein. The label shall include the following information (see 4.4.1):

Intermediate Cold Flyer's Glove, HAU-15/P
Name of manufacturer
Contract or Order No.
NSN
Size

3.5. Instruction slip. A printed slip of thin paper containing instructions as indicated below shall be inserted in each right hand glove.

Intermediate Cold Flyer's Glove
Instructions

This glove is intended for use in intermediate cold, dry conditions. To maintain maximum dexterity and tactility, the insulation on the palm and palm side of fingers is lighter than that on the back of the hand. The glove shell and knit cuff are made from flame resistant Nomex fiber, with sheepskin leather used for the palm.

The gloves may be hand washed with mild soap in warm water, not over 120°F. Rinse thoroughly. Wrap in a towel to remove excess moisture. Do not wring or twist. Air dry, without exposure to hot air or full sunlight.

3.6 Design. The gloves shall be a two layer, slip-on, inseamed Flexor® Design with a straight thumb, knitted cuff, seamless fingertips, reinforcement patch across the palm and thumb crotch, and elastic take-ups at the wrist and across the back of the hand (see 6.6 and Figures 1 & 2).

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3.7 Patterns . The government will furnish a complete set of patterns. The government pattern shall not be altered in any way and shall be used to create the contractor's working pattern. Minor modifications are permitted to the contractor's working pattern or dies where necessary when using automatic equipment or production processes. These modifications shall not alter the appearance, serviceability or dimensional requirements cited in this document.

3.7.1 List of pattern parts The pattern parts and cutting quantities for the gloves shall be as listed in Table V (see 6.7).

TABLE V. List of pattern parts

Basic Material	Nomenclature	Cutting Quantity per pair
Shell Leather	Finger palm: index & middle fingers (left and right)	2
	Finger palm: ring and little fingers (left & right)	2
	Hand palm (left and right)	2
	Palm patch (left & right)	2
	Back Welt	2
Shell, Knit Cloth	Finger back (left & right)	2
	Thumb back (left & right)	2
	Hand back (left & right)	2
Lining A	Finger back (left & right)	2
	Thumb back (left & right)	2
	Hand back (left & right)	2
Lining B	Finger palm (left & right)	2
	Hand palm (left & right)	2

3.8 Construction.

3.8.1 Stitches, seams and stitching. Stitch, seam, and stitching types specified in Table VI shall conform to ASTM D6193. Seam allowances shall be maintained with seams sewn so that raw edges twists, pleats, or puckers will not result. All seams shall start and finish evenly. Ends of a continuous line of stitching shall be overlapped not less than 1/2 inch.

3.8.1.1 Types 301 and 304 stitching. Ends of 301 seams and stitching, when not caught in other seams or stitching, shall be bartacked not less than 1/4 inch. Ends of 304 stitching shall be caught in the closing seam of the glove. Thread tension shall be maintained so that there will be no loose tension resulting in loose bobbin or top thread, or excessively tight stitching resulting in puckering of the material sewn.

3.8.1.1.1 Repairs of type 301 stitching. Repairs of stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch of the end of the stitching. The ends of the stitching are not required to be backstitched.

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b. Thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch back of the defective area, continue over the defective area, and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the material and restitching in the required manner. The ends of the stitching are not required to be backstitched.

3.8.1.1.2 Repairs of type 304 stitching. Repairs of stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of three stitches back of the end of the stitching. The ends of the stitching are not required to be backstitched.
b. Thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of three stitches back of the defective area, continue over the defective area, and continue a minimum of three stitches beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the material and restitching in the require manner. The ends of the stitching are not required to be backstitched.

3.8.1.2 Type 401 stitching. Thread tension shall be maintained so that there will be no loose or excessively tight stitching. All repairs shall be made in accordance with 3.8.1.1.1a and 3.8.1.1.1b. Repairs to type 401 stitching may be accomplished by use of stitch type 301.

3.8.1.3 Type 503 or 504 stitching. Thread tension shall be maintained such that when the two pieces of base fabric joined by the stitching are gently pulled perpendicular to the seam, the needle thread shall be barely visible and there shall be no visible spaces between the stitches, when viewed from the face side of the assembly. No sign of puckering shall be observed in the materials sewn.

3.8.2 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in Table VI.

3.9 Manufacturing operations requirements. The gloves shall be manufactured in accordance with the operation requirements specified in Table VI. The contractor is not required to follow the exact sequence of operations listed. Any additional basting or holding stitching used to facilitate manufacture is permissible provided the thread is removed or does not show in the finished gloves.

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TABLE VI. Table of Operations

NO	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	NEEDLE THREAD	BOBBIN/ LOOPER THREAD
1.	Cutting a. The skins may be dampened with tap water and excess stretch shall be removed by staking, toggling or other means.					
	b. With the exception of finger pieces, all fabric portions (shell and lining) shall be cut in accordance with the patterns. If no directional line is given on the pattern the long dimension of the piece is cut in the direction of the fabric warp. Finger patterns shall be cut on the bias.					
	c. All leather pieces except the leather welt shall be die cut. The welts may be hand cut. The width shall be 3/16 ± 1/16 inch. The welt length shall be sufficient in length to extend within 1/8 inch or less of each side seam in the finished glove.					
2.	Replacement of damaged parts. During the spreading, cutting, and manufacturing operations components having material defects or damages, as classified in 4.4.3, shall be excluded and replaced with non-defective and properly matched components. Repairs such as mends, patches, or splices are not permitted.					

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TABLE VI. Table of operations - Continued

NO	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	NEEDLE THREAD	BOBBIN/ LOOPER THREAD
3.	Marking. a. Mark, ticket, or bundle all cut parts to ensure a uniform size throughout each glove. b. Any method of marking shall be used except: (1) Metal fastening devices. (2) Sew on tickets. (3) Adhesive type tickets which leave traces of adhesive on the material after removal of the ticket.					
4.	Construct knit outer shell fingers. a. Fold each finger on the second notch closest to the fingertip. Align the folded edge even with the lower notch forming a pleat not less than 1/4 + 1/16, -0 inch in width, folded toward the hand. Tack the pleats 1/16 + 1/16, -0 inch from edge (see B of figure 3). Start the stitching not less than 1/8 inch before the pleat and extend it at least 1/8 inch beyond the pleat.	301 (reverse double stitch) or bartack	OSE -1	9-11	B	B
	b. Open the finger back (star). The inside square will form a straight line. Align the finger back and hand back (see A of figure 3). The notch on the hand back indicates the little finger side of the piece. The notches on the finger back are placed on the little finger of	301	SSj-1	9-11	B	B

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TABLE VI. Table of operations - Continued

NO	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	NEEDLE THREAD	BOBBIN/ LOOPER THREAD
4.	Construct knit outer shell fingers. (cont'd) the piece. They are not intended to match. Insert and center the welt between the two pieces. Stitch all three layers together 1/8 ± 1/16 inch from edge. (See A of figure 1).					
5.	Construct leather outer shell finger. a. Join together the two pieces of the finger palm (star) between the middle and ring fingers with a lapped seam 1/2 ± 1/16 inch wide. Align the "V" shapes at the inside of the star and between the fingers. Stitch together with two rows of stitching, 1/8 inch apart, 1/4 ± 1/16 inch from raw edge.	301	LSa-2	9-11	B	B
	b. Open the finger palm (star). The inside square forms a straight line. Lap the finger palm over the hand palm. Align the notches on the hand palm. Stitch together with 1/8 ± 1/16 inch from edge (see A of figure 3).	301	LSa-1	9-11	B	B
	c. Join the thumb back to the thumb palm at the thumb tip with a 1/4 ± 1/16 inch seam. Align the raw edge of each piece with the notches on the opposite piece. The thumb tip shall be lapped over the thumb back. Stitch together 1/8 ± 1/16 inch from edge.	301	LSa-1	9-11	B	B

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TABLTE VI. Table of operations - Continued

NO	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	NEEDLE THREAD	BOBBIN/ LOOPER THREAD
	d. Align front of thumb back with the hand palm and stitch together 1/8 ± 1/16 inch from edge. Do not stitch across tip of thumb.	301	SSa-1	9-11	B	B
	e. Align back of thumb back to hand palm and stitch together 1/8 ± 1/16 inch from edge. Do not stitch across tip of thumb.	301	LSa-1	9-11	B	B
	f. Position palm patch on palm by aligning the bottom corner (opposite the thumb) with the notch on the little finger side of the palm. Stitch the patch along the top edge, smoothing out wrinkles. Tack about 1 inch along the edge on the thumb side. Smooth the patch over the palm and sew along the bottom edge and around the thumb area. The peninsula of the palm patch extends down the back side of the thumb (see figure 2). Stitch palm 1/8 ± 1/16 inch from edge.	301	LSa-1	9-11	B	B
	g. Join the hand back to the palm assembly at each fingertip 1/4 ± 1/16 inch from edge. Align the raw edge of each piece with the notches on the opposite piece. The fingertips shall be lapped over the hand back. Stitch together 1/8 ± 1/16 inch from edge.	301	LSa-1	9-11	B	B
6.	Attach elastic webbing. a. Elastic webbing, of the lengths indicated in table IV, shall be attached across the hand back, at mid-palm and at the wrist, as indicated by the notches.	304	SSaa-1	9-11	B	B

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TABLE VI. Table of operations – Continued

NO	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	NEEDLE THREAD	BOBBIN/ LOOPER THREAD
6.	Attach elastic webbing. (cont'd) a. The elastic shall be zig-zagged stitched to the inner side of the pieces, positioning the fully extended elastic across the pieces and aligning the center of the elastic width to the notches on the sides of the pieces. Stitch through the elastic and leather or fabric. The fullness of the glove shall be distributed evenly under the relaxed elastic.	304	SSaa-1	9-11	B	B
	b. Elastic webbing, of the lengths indicated in table IV, shall be attached across the thumb back and palm, at the wrist, as indicated by the notches. The elastic shall be zig-zagged stitched to the inner side of the pieces, positioning the fully extended elastic across the pieces and aligning the center of the elastic width to the notches on the sides of the pieces. Stitch through the elastic and leather or fabric. The fullness of the glove shall be distributed evenly under the relaxed elastic.	304	SSaa-1	9-11	B	B
7.	Assemble shell back and palm. a. Align the back assembly with the palm assembly. Close shell by stitching together from edge 1/8 (+1/16, -0) inch at the fingertip, with the back assembly on top, starting at either the little finger side (right glove) or	301	SSa-1	9-11	B	B

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TABLE VI. Table of operations -Continued

NO	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	NEEDLE THREAD	BOBBIN/ LOOPER THREAD
7.	Assemble shell back and palm (cont'd) thumb side (left glove). Do not sew across the finger and thumb tips. Take care to sew to the inboard side of the finger pleat tacks and catch the ends of the threads used in joining the fingers to the hand pieces. The sides of the palm patch shall be caught in the closing seam. The elastic webbing on the back of the hand and at the wrist and the size tag stitching shall be fully caught in the closing seam.	301	SSa-1	9-11	B	B
	b. Turn hem 1/4 ±1/16 inch with backs sides of the material together and stitch 1/8 ±1/16 inch from turned edge.	301	EFa-1	9-11	B	B
8.	Lining A hand and finger back. a. Open the finger back (star). The inside square will form a straight line. Align it with the hand back, and stitch together, 1/8 ± 1/16 inch from edge, leaving a thread tail on both ends. The notch on the hand back indicates the little finger side of the piece. The notches on the finger of the piece. They are not intended to match.	401	SSa-1	8-10	B	B
9.	Lining B, hand and finger a. Open the finger palm (star). The inside square will form a straight line. Align it with the hand palm, and stitch together, 1/8 ± 1/16 inch from edge, leaving a thread tail on both ends.	401	SSa-1	8-10	B	B

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TABLE VI. Table of operations - Continued

NO	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	NEEDLE THREAD	BOBBIN/ LOOPER THREAD
10.	Thumb. a. Align the thumb back with the thumb tip and stitch together, 1/8 ±1/16 inch from edge, leaving a thread tail on both ends.	401	SSa-1	8-10	B	B
	b. Align the thumb back with the hand palm and stitch together, 1/8 ± 1/16 inch from edge, leaving a thread tail on both ends. Do not stitch across the tip of the thumb.	401	SSa-1	8-10	B	B
11.	Assemble lining back and palm. a. Align the hand back with the palm assembly at each fingertip and stitch together, 1/8 ± 1/16 inch from edge, leaving a thread tail on both ends.	401	SSa-1	8-10	B	B
	b. Align the back assembly with the palm assembly and stitch together, 1/8 ± 1/16 inch from edge, leaving a thread tail on both ends. Catch the ends of the threads used in joining the fingers to the hand at the palm and fingertips. Do not stitch across the fingertips.	401	SSa-1	8-10	B	B
12.	Attach lining to shell. a. Align fingertips of lining to fingertips of shell with palm side together such that the fingertips of the lining are even with the fingertips of the shell. Using at least 5 stitches per finger, stitch the lining to the shell at one side of each finger and thumb tip by	301	SSa-1	9-11	B	B

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TABLE VI. Table of operations - Continued

NO	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	NEEDLE THREAD	BOBBIN/ LOOPER THREAD
12	Attach lining to shell. (cont'd) sewing the tab on the side of the lining tips to the side of the shell tip, taking care to place the stitching inboard of the shell closing seam.					
	b. Turn the shell over the lining, taking care that the shell fingertips are fully opened with the ends well rounded, and that the lining fingertips are fully extended into the shell fingertips.					
13.	Attach cuff. Align the knit cuff to the edge of the lining and stitch together outseam, 1/8 ± 1/16 inch from edge, catching label in the seam. The top edge of the label shall be inserted in the seam at center palm, on the interior of the glove, with the printed size displayed.	503, 504	SSa-1	9-11	B	B
14.	Hem glove. Turn the hem of the outer shell 3/8 to 1/2 inch over the knit cuff seam and stitch closed, 1/4 ± 1/16 inch from the folded edge of the hem.	301	SSbc-1	9-11	B	B
15.	Finishing. a. The gloves shall be laid off properly shaped an appropriately sized electrical hand forms heated to 175^0 to 200^0F.					
	b. Trim all threaded ends to not more than 1/8 inch.					
	c. Clean gloves by removing all lint, loose threads, and shade and size markings.					

3.10 Figure Figures 1,2 and 3 are furnished for information purposes only. If there are any inconsistencies between the specification and the figures, the specification shall govern.

3.11 Finished lining measurements. This in-process measurement shall be taken to assure

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that the proper tension is being applied when stitching the lining. The finished measurements for the lining shall be shown in Table VII.

TABLE VII. Finished lining measurements (inches)

SIZE	WIDTH	TOLERANCE
5	4 -7/16	+/-1/16
6	4 -9/16	+/-1/16
7	4 -11/16	+/-1/16
8	4 -13/16	+/-1/16
9	4 –15/16	+/-1/16
10	5-1/16	+/-1/16
11	5-3/16	+/-1/16

NOTE: The lining shall be laid flat on a table & measurements shall be taken from stitching at the seam joining the lining palm and lining handback to the fingers.

3.11.1 Finished glove measurements (overall) The finished measurements shall be as shown in Table VIII. Measurements shall be made using a 1/4 inch wide metal measuring tape which has, at the beginning of the tape, a metal tip positioned at a right angle to the tape. The metal tip shall be hooked over and placed against the edge of the glove when taking measurements. Measurements shall be to the nearest 1/8 inch.

TABLE VIII Finished glove measurements, overall (inches)

SIZE	LENGTH B	WIDTH C	TOLERANCE
5	9-1/2	4 13/16	+1/16, -1/8
6	10	5 1/16	+1/16, -1/8
7	10-1/2	5 3/16	+1/16, -1/8
8	11	5 5/16	+1/16, -1/8
9	11-1/2	5 9/16	+1/16, -1/8
10	12	5 9/16	+1/16, -1/8
11	12-1/2	5 9/16	+1/16, -1/8

Note: (Refer to Figure 1 for the B and C locations)

B. length

The glove length shall be measured from the tip of the middle finger to the edge of the glove body, i.e., the folded edge of the simplex or leather. Do not include the knit cuff in the length measurement. The glove shall be gently stretched lengthwise and then allowed to return to its natural state before measuring.

C. Width.

The glove width shall be gently stretched out and measured on the palm, 1/2 inch from the base of the little finger.

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3.12 Finished measurement across folded area of fingertips. The finished fingertip measurement across any finger or thumb shall be not less than 1/2 inch and not more than 15/16 inch. Measurements shall be made with a ruler having 1/16 inch increment. Measurements shall be made by holding an individual finger upright and abutting the ruler to the top edge of the fingertip (See Figure 1, D)

3.13 Workmanship The finished gloves shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Conformance inspection (see 4.3)

4.2 First article inspection. When a first article is required (see 3.1), it shall be examined for the requirements of Table IX.

TABLE IX. First Article examinations and tests.

Inspection	Requirement	Inspection Method
Component and Material	3.3. 1, 3.3.2, 3.3.3.1,	4.4.1
Inspection	3.3.3.2, 3.3.3.3, 3.3.3.4	
	3.3.4, 3.3.5, 3.3.6, 3.4	
Visual Examination	3.6, 3.8.1, 3.8.1.1, 3.8.1.2, 3.8.1.3, 3.8.2, 3.9, 3.13	4.4.3
Dimensional Examination	3.11, 3.11.1, 3.12	4.4.4

4.2.1 First article units. The pairs of gloves selected as sample units from the first article lot shall be thoroughly checked for dimensional and visual conformance to this specification. The sample size of the first article lot shall be specified in the contract. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. The first article inspection samples shall be furnished to the Government as directed by the contract (see 6.2).

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4.3 Conformance inspection. Unless otherwise specified, conformance testing shall be performed on production lot articles for the requirements of Table X. Sampling for inspection shall be performed in accordance with ANSI/ASQC Z1.4 (see 6.2).

.TABLE X. Conformance Inspection examinations and tests

Inspection	Requirement	Inspection Method
Visual Examination	3.6, 3.8.1,3.8.1.1, 3.8.1.2,	4.4.3
	3.8.1.3, 3.8.2, 3.9, 3.13	
Dimensional Examination	3.11, 3.11.1, 3.12	4.4.4

4.4 Inspection methods.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with the requirements of Table XI.

TABLE XI. Component Parts

Material	Characteristic	Paragraph	Test Method
Leather	All Applicable under	3.3.1	All Applicable under
	MIL-DTL-32067		MIL-DTL-32067
Cloth	All Listed Under	3.3.2	All listed under
	MIL-C-81393		MIL-C-81393
Lining: Backing and	Material identification	3.3.3.1	All listed under
Facing			
	Color	3.3.3.1	All listed under
	Weight	3.3.3.1	ASTM - D-3776, option C1/
	Fabric Count	3.3.3.1	ASTM - D-3887 1/
	Bursting Strength	3.3.3.1	D-3787 1/
Polyester Insulation	Material Identification	3.3.3.2	All tests specified in MIL-B-44346
Lining A			
	Thickness	3.3.3.2	ASTM - D-1777 1/3/
Polyester Insulation Lining B	Weight Material Identification	3.3.3.2 3.3.3.3	ASTM - D-3776 1/4/ All tests specified in MIL-B-44346
	Weight	3.3.3.3	ASTM – D-3776 1/4/

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TABLE XI. COMPONENT PARTS

Material	Characteristic	Paragraph	Test Method
Finished Lining Laminates	Thickness	3.3.3.4	ASTM-D-1777 1/3
	Bond Strength	3.3.3.4	ASTM-D-2724, 3/4/
	Drape Stiffness	3.3.3.4	ASTM-D-1388, option A 2/
Knit Cuff	Material Identification	3.3.4	All tests specified in MIL-C-3735 visual
	Length	3.3.4	All listed under
Elastic	Material Identification	3.3.5	All tests specified in MIL-W-5654
Thread	Material Identification	3.3.6	All Tests Specified in A-A-50195, V-T-295 A-A-50199 as applicable
Label	Material Identification	3.4	All Tests specified in MIL-DTL-32075

1/ Refers to ASTM
2/ Determination of leather thickness shall be accomplished by measuring a folded sample area and dividing by two.
3/ Measured using a thickness gauge with a 12 by 12 inch pressure foot applying a load of 0.002 lbs/sq in.
4/ Six 12 by 12 inch square specimens shall be tested. The individual and mean weight of the six specimens shall be reported to the nearest 0.1 ounces per square yard.

4.4.2 In-process inspection. Inspection of sub-assemblies shall be made to ascertain that construction details and the lining width which cannot be examined in the finished product are in accordance with specified requirements. The gloves shall be examined after stitching of the elastic to assure the length of the elastic from the edge of the material to the opposite edge of the material conforms to Table IV. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated non-conformance.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in Table XII. The lot size shall be expressed in units of gloves. The sample unit shall be one glove and selection shall be by pairs. Defects for pairing shall be classified as a single defect.
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TABLE XII. End Item Visual Defects

Examine	Defect	Classification Major	Minor
Pairing	Right and Left not same size	101	
Color	Leather portions of shell not black	102	
	Fabric portions of shell, including cuff, not sage green	103	
	Lining color not white		201
	Leather portions of shell not of uniform color		202
	Fabric portions of shell, including cuff, not of uniform color		203
Shell or lining design	Not as specified (incorrect material, pattern, etc.)	104	
Cleanness	Spots or stains clearly noticeable affecting appearance		204
Leather	Hole (except a pinhole or needle hole), cut, tear or rip	105	
	Three or more unspecified pinholes or needle holes		205
	Not full grain	106	
	Not soft and flexible	107	
	Grain damage or imperfections: seriously affecting serviceability or appearance	108	

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TABLE XII. End Item Visual Defects - Continued

Examine	Defect	Classification Major	Minor
	Affecting appearance (not applicable to minute surface break providing the condition of frequency of same does not have serious effect on appearance)		206
	Deep open scratch	109	
	Deep wrinkles seriously affecting serviceability or appearance	110	
	Wrinkles affecting appearance		207
	Scar or abraded area: - seriously affecting serviceability or appearance	111	
	- affecting appearance		208
	Brittle, bony area, or thin spot Seriously affecting serviceability:	112	
	- affecting serviceability, but not seriously		209
	Loose or spongy leather	113	
	Flanky leather: - seriously affecting serviceability or appearance	114	
	Affecting serviceability but not seriously		210
Lining	Facing delaminated		211
	Any hole, cut, tear or rip.	115	
Knit cuff	Any fabric defects such as or snags or runs		
	- seriously affecting serviceability or appearance	116	

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TABLE XII End Item Visual Defects - Continued

Examine	Defect	Classification Major	Minor
	Affecting serviceability or appearance, but not seriously		212
	Any hole, cut, tear, or rip	117	
Construction, and workmanship; general (applicable to all components unless otherwise indicated herein)	Component misplaced, operation omitted, or operation not properly performed	118	
	Lining reversed so that nylon tricot backing is oriented toward hand	119	
	Component missing or not securely affixed	120	
	Mend in leather or fabric (i.e, patch not applicable to restitched seam repair)	121	
	Needle chews likely to develop into a hole	122	
Seams and Stitching	Open seam: - In single stitched seam - In any row of a double stitched seam -not repaired as specified (when applicable)	 123 124	 213

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TABLE XII. End Item Visual Defects -Continued

Examine	Defect	Classification Major	Minor
	NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped or run-off stitches occur.		
	Not specified seam type	125	
	Not specified stitch type	126	
	Row of stitching omitted	127	
	Loose stitch tension resulting in a loosely secured seam	128	
	Tight stitch tension resulting in cutting of leather or fabric or breaking of stitches when normal pull is applied	129	
	Ends of stitching not secured as specified		214
	Gage of stitching irregular or not as specified, affecting Appearance	130	
	Part caught in unrelated row of stitching	131	
	One or two stitches per inch less than specified		215
	Three or more stitches per inch less than specified	132	
	Leather or fabric torn by needles (elongated holes) - up to 1/2 inch - greater than 1/2 inch	 133	216

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TABLE XII End Item Visual Defect Continued

Examine	Defect	Classification Major	Minor
	Not backstitched where required	134	
	More than the specified maximum number of stitches per inch:		
	- damaging materials	135	
	- not damaging materials		217
Assembly Detail	Glove not neatly laid off:		
	- wrinkles in shell		218
	- fingers not completely opened	136	
	No inseam construction where specified	137	
	Finished measurement across folded area of any fingertip less than specified:		
	- up to 1/16 inch less		219
	- greater than 1/16 inch less	138	
	Welt omitted on back	139	
	Welt is more than 1/8 inch from either side seam		220
	Welt not completely caught in finger back/hand back seam		221
	Welt caught in closing seam		222
	Pleats missing from any finger (does not include thumb)	140	
	Shell more than 1/8 inch longer than lining in any finger when on hand or on flat surface	141	

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TABLE XII End Item Visual Defects -Continued

Examine	Defect	Classification Major	Minor
	Poorly assembled seriously affecting serviceability (e.g., finger distorted or twisted affecting comfort of wearer or causing seam finger to come in contact with object)	142	
	NOTE: Any construction defect that results in a malformed glove such as a twisted finger or thumb resulting in any seam lying in the palmar area of the finger or thumb is a major defect. When a finger or thumb is twisted and the seam is rotated to less than 45 degrees from the vertical and does not lie in the palmar area, it is a minor defect		
	Poorly assembled affecting serviceability but not seriously (e.g., pleats on palm or thumb back created when attaching palm patch, fabric not distributed evenly under elastic take-ups, wrist elastic misaligned at side seams etc.)		223
	Edge of pleat not fully caught in closing seam - less than 1/16 inch - more than 1/16 inch	 143	224
	Lining not securely fastened to shell at side of fingertips	144	
	Knit cuff seam not caught in cuff hem	145	

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TABLE XII End Item Visual Defects - Continued

Examine	Defect	Classification Major	Minor
	Thread ends not trimmed throughout glove		225
	Not constructed with specified number of pieces	146	
	Difference in overall length between front and back of glove is more than 1/2 inch	147	
	Elastic webbing across back of Hand and at wrist not fully caught in closing seam	148	
Label and Instruction Slip	Omitted, incorrect, illegible, or misplaced; size and identification label not securely sewn in seam		226

4.4.4 End item dimensional examination. The end item shall be examined for conformance to the dimensions specified in Table VIII, 3.11 and 3.12. Any dimension not within the specified tolerance shall be classified as a Major defect. The defect number shall be 149. The lot size shall be expressed in units of gloves. The sample unit shall be one glove and the selection shall be by pairs.

5. PACKAGING

5.1 Packaging. For Acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contract the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity."

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The glove covered by this Specification is intended to be worn by military aircrew requiring an intermediate level of cold weather protective handwear.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Sizes required (see 1.3).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
d. When first article is required (see 3.1, 4.2 and 6.3).
e. Packaging requirements (see 5.1).

6.3 First article. When a first article is required, it will be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Patterns. For access to patterns, address the acquiring activity issuing the invitation for bids.

6.5 Source of supply

6.5.1 Lining fabric. Finished lining fabric meeting the requirements of 3.3.3.4 is available from:

International Foam Products, Inc.
771 Commercial Avenue
Carlstadt, NJ 07072

6.5.2 Knit cuffs. Knit cuffs meeting the requirements of 3.3.4 are available from:

St. Regis Sportswear
51 Beaver Street
Gloversville, NY 12078

6.6 Patent notice. U.S. Patent number 4654896 covers the Flexor design of this item as described in 3.6 of this specification. The Government does not have a royalty-free license for this patent. Information regarding licensing for production of gloves meeting this specification may be obtained from:

Rinehart Glove Ltd.
P.O. Box 3400
Montrose, CO 81401

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6.7 Pattern Numbers. The patterns listed in 3.7.1 have been identified with a NAVAIR drawing number for the purpose of configuration control and tracking of pattern changes. These numbers range from 3526AS190-(1 to 13) through 3526AS196-(1 to 13). These numbers do not have any effect on manufacturing.

6.8 Subject terms (key word) listing.

Cold Weather Gloves

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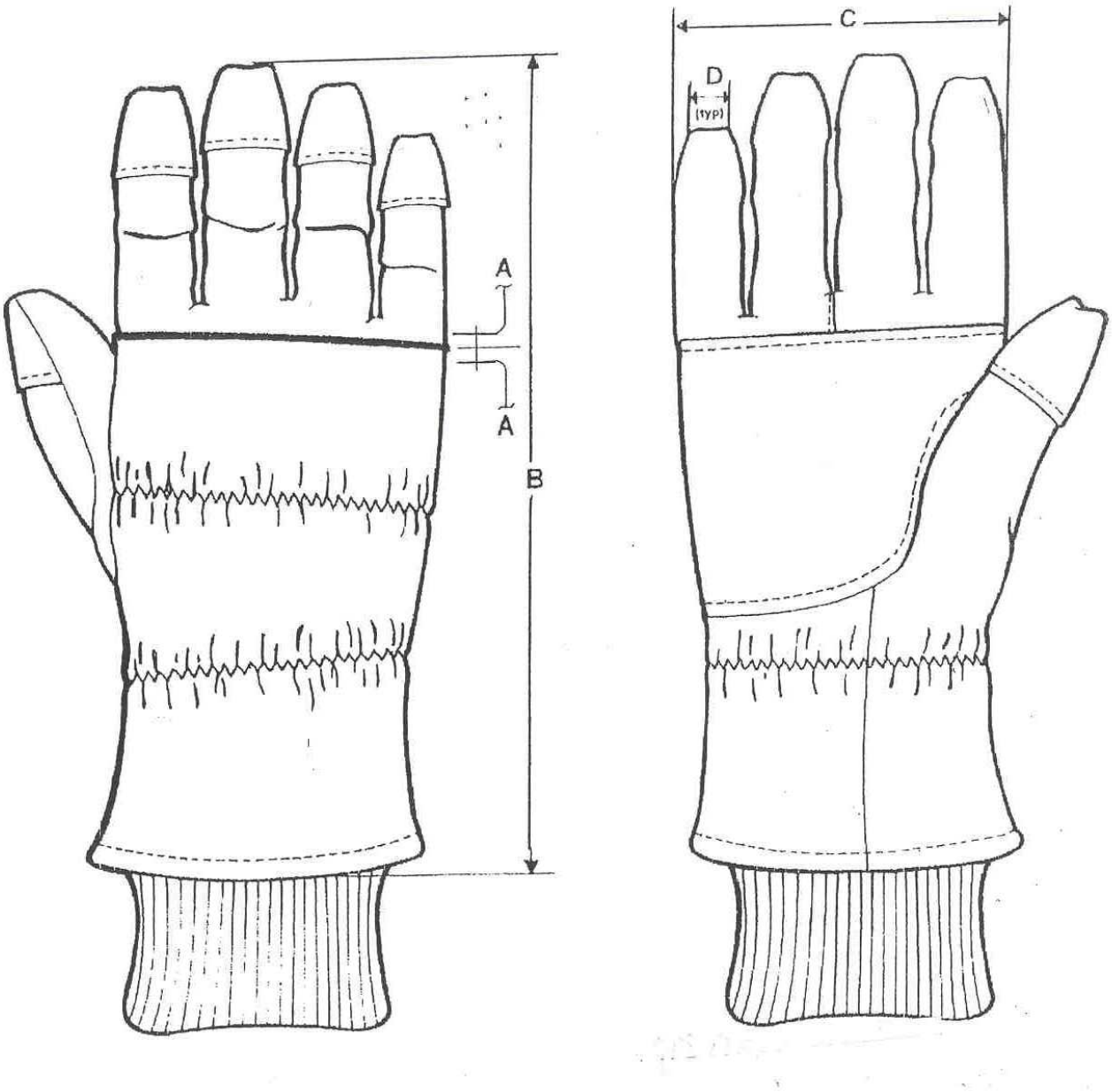


Figure 1 HAU-15/P Glove

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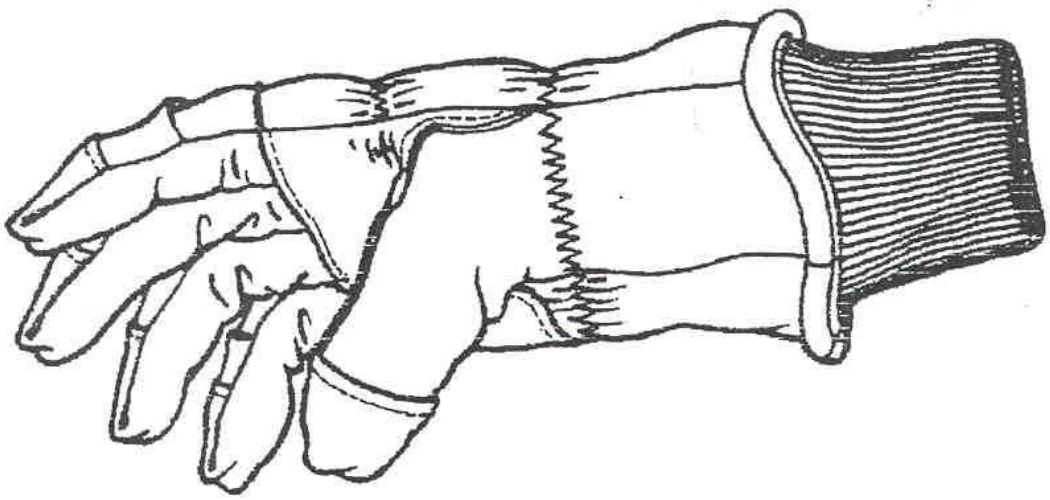


Figure 2 HAU-15/P Palm Patch Position

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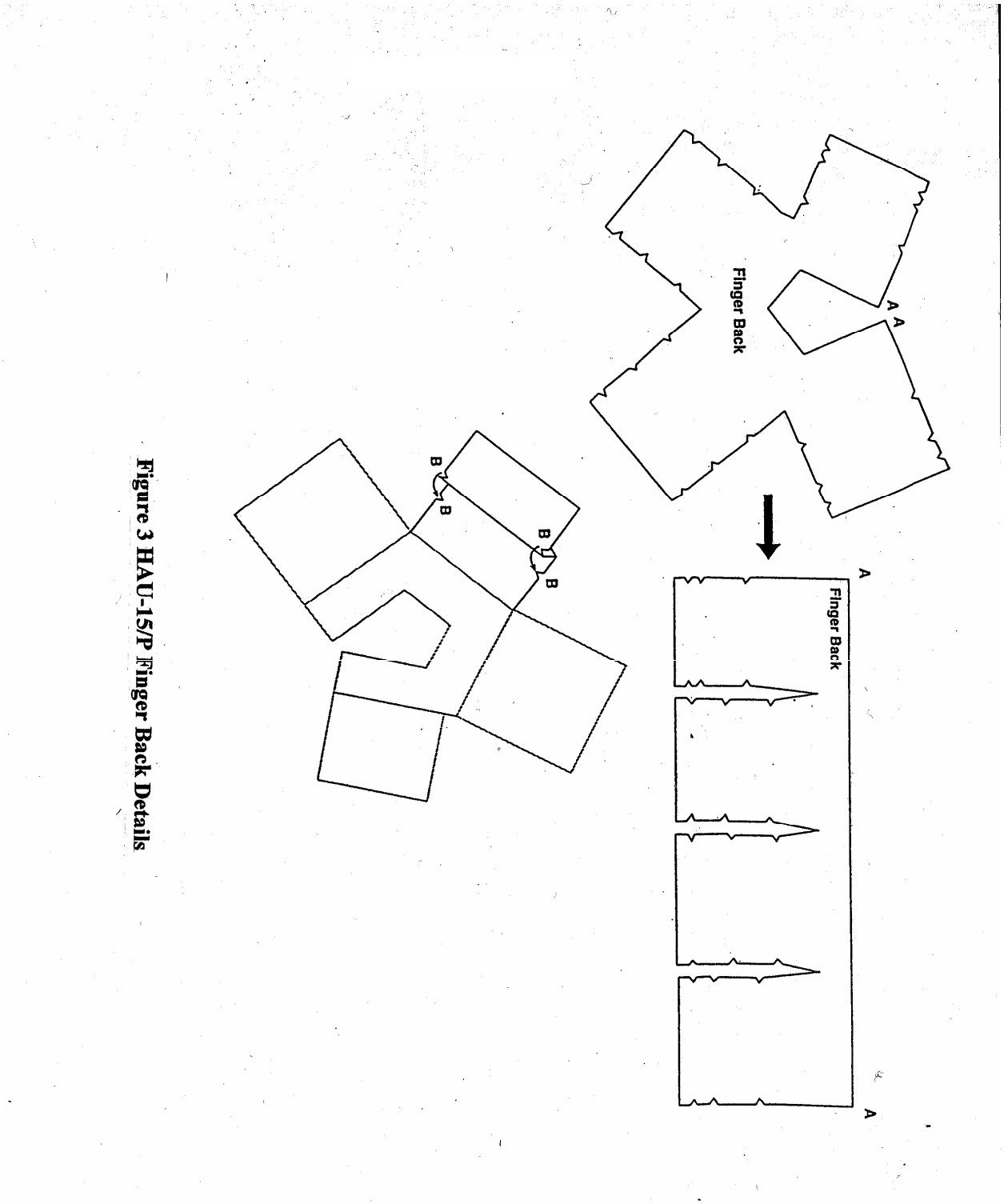


Figure 3 HAU-15/P Finger Back Details

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Custodians:

Navy - AS
Army -GL
Air Force 11

Preparing activity:

DLA-CT

Review Activity:

Navy – NU

Project Number:

8415-0262

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